



Phylogeny and biogeography of the *Poecilia sphenops* species complex (Actinopterygii, Poeciliidae) in Central America

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ARTICLE INFO

Article history:

Received 23 May 2012

Revised 6 November 2012

Accepted 15 December 2012

Available online 28 December 2012

Keywords:

Incomplete lineage sorting

Mesoamerica

Molecular clock

Molecular systematics

Poecilia

Secondary freshwater fishes

ABSTRACT

We inferred the phylogenetic relationships among members of the *Poecilia sphenops* species complex to resolve the colonization process and radiation of this group in Central America. We analyzed 2550 base pairs (bp) of mitochondrial DNA (mtDNA), including ATP synthase 6 and 8, cytochrome oxidase subunit I and NADH dehydrogenase subunit 2 genes, and 906 bp of the nuclear S7 ribosomal protein of 86 ingroup individuals from 61 localities spanning most of its distribution from Mexico to Panama. Our mitochondrial data rendered a well-supported phylogeny for the *P. sphenops* complex that differed with the nuclear data set topology, which did not recover the monophyly of the *P. mexicana* mitochondrial lineage. Coalescent-based simulations tests indicated that, although hybridization cannot be completely ruled out, this incongruence is most likely due to incomplete lineage sorting in this group, which also showed the widest geographic distribution. A single colonization event of Central America from South America was estimated to have occurred between the early Paleocene and Oligocene (53–22 million years ago). Subsequently, two largely differentiated evolutionary lineages diverged around the Early Oligocene–Miocene (38–13 million years ago), which are considered two separate species complexes: *P. sphenops* and *P. mexicana*, which can also be distinguished by their tricuspid and unicuspid inner jaw teeth, respectively. Ultimately, within lineage diversification occurred mainly during the Miocene (22–5 million years ago). All major cladogenetic events predated the final closure of the Isthmus of Panama. The allopatric distribution of lineages together with the long basal internodes suggest that vicariance and long term isolations could be the main evolutionary forces promoting radiation in this group, although dispersal through water barriers might also have occurred. Lastly, our results suggest the need to review the current species distribution and taxonomy of the *P. sphenops* complex *sensu lato*.

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1. Introduction

Central America (CA) has a highly complex geological history that has strongly influenced the historical and current distributions of its fauna. Therefore, patterns of species' occurrence have long been studied in the context of the geological and ecological history of CA (Savage, 1966; Bussing, 1985; Webb, 1991; Smith and Bermingham, 2005). In this region, freshwater fish diversity is assumed to be low in comparison to the species diversity found in South America (SA), where most of the ichthyofauna is originary (Myers, 1966; Bussing, 1985). Different authors have stated that this poverty of freshwater fishes must be mainly due to the young age of CA, and to their low dispersal capability. Hence freshwater fish distribution is largely dependent on connections between drainage basins, thus implying a significant interplay between bio-

logical and geological evolution (Miller, 1966; Bussing, 1985; Bermingham and Martin, 1998; Hulsey and López-Fernández, 2011). In the case of explaining the Neotropical origin of CA ichthyofauna, it is typically assumed that fishes arrived to CA through a freshwater connection between continents. However, since the Late Cretaceous–Early Paleocene no land connection has been proposed to exist between the two landmasses until the final closure of the Isthmus of Panama about 3–3.5 Mya (Coates and Obando, 1996; Iturralde-Vinent and MacPhee, 1999; Iturralde-Vinent, 2006; but see Montes et al., 2012). According to this, freshwater fishes could not have colonized CA until the Pliocene (Bussing, 1985). Therefore, contemporary patterns of freshwater fish distribution in CA must be explained either by dispersal through the sea or over a land bridge (Bermingham et al., 1997b). However, there are a number of examples of fishes that have colonized CA long before the closure of the Isthmus (Martin and Bermingham, 1998; Murphy et al., 1999; Perdices et al., 2005; Concheiro Pérez et al., 2007). Most of these examples are characterized by secondary freshwater fishes, which have a close marine ancestor and are thus tolerant to some extent to saline or brackish waters, although some CA

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lineages of primary freshwater fishes and amphibians also predate the closure of the Isthmus (Perdices et al., 2002; Ornelas-García et al., 2008; Pinto-Sánchez et al., 2012).

In addition to the earlier colonization of secondary freshwater fishes in CA, these fishes are highly diverse and, in contrast to SA, they dominate the freshwater fish fauna in CA (Myers, 1966; Bussing, 1976, 1985). It is proposed that the early arrival to a continent without competitors (“ostariophysan vacuum” *sensu* Myers, 1966) could have provided an advantage for the large and successful diversification of this group of fishes (Betancur et al., 2012). Indeed, some secondary freshwater fishes are classic examples and models for the study of biological diversification in CA. For example, heroine cichlids have undergone extraordinary radiations leading to a large number of species with wide morphological variations usually related to their ecological adaptations (Barluenga et al., 2006; Elmer et al., 2009; Muschick et al., 2011). Poeciliids, although not so thoroughly studied as cichlids, are also secondary freshwater fishes, widely distributed across the American continent and representing a large radiation (Brett and Turner, 1983). Yet, the processes leading to the high diversity within the genus *Poecilia* are largely unknown. In the present work, we focus on the *Poecilia sphenops* species complex, a group of species of the family Poeciliidae occurring from Mexico to Panama (Miller, 2005).

Unlike other fishes with a SA origin, poeciliids in CA usually constitute the dominant species where they occur and are highly diverse, accounting for approximately 35% of CA secondary freshwater fish fauna (Miller, 1966). Because of this abundance and conspicuity, poeciliids have played a key role in studies of vicariance biogeography of CA and the Caribbean (Rosen, 1975; Bussing, 1976; Rivas, 1986; Rauchenberger, 1988; Hrbek et al., 2007). Bussing (1976) considered this genus to be member of an “old southern element” that, together with other genera (i.e., *Cichlasoma*, *Brachyrhaphis*, and probably *Rhamdia*), reached CA in the Late Cretaceous or Paleocene through an intercontinental land bridge. However, *Poecilia* extended its range further north, experiencing its largest diversification in upper CA, while fewer allopatric species occur in lower CA (Bussing, 1976). A more recent hypothesis based on molecular data also supports an ancient pre-Cretaceous origin of poeciliids, and proposes the existence of multiple colonizations from SA (Hrbek et al., 2007). *Poecilia* would thus constitute a third wave of colonization, which together with its sister genus *Limia* dispersed to the Greater Antilles and CA in the early Miocene (Hrbek et al., 2007). Therefore, according to the hypotheses mentioned above, we would expect that all *Poecilia* in CA share a common ancestor, which dispersed from SA long before the closure of the Isthmus of Panama 3–3.5 Mya.

Central American *Poecilia* is considered by some authors, based on morphological characters of the gonopodium, as the group or subgenus *Mollienesia*, also known as mollies (Rosen and Bailey, 1963; Miller, 1975; Rauchenberger, 1989), although some studies do not find enough morphological evidence for such separation (Poeser, 2003). On the other hand, molecular data support the monophyly of this group (Ptacek and Breden, 1998). Given this disagreement, here we will follow the most recent nomenclature based on generic names (Lucinda, 2003; Lucinda and Reis, 2005). Within CA *Poecilia*, two readily distinguishable, and reciprocally monophyletic (Ptacek and Breden, 1998), species groups are found: the long “sail” fins of the *P. latipinna* complex and the short dorsal fins of the *P. sphenops* complex (Hubbs, 1926; Schultz and Miller, 1971; Miller, 1983), but aside from this character, the genus and particularly short-fins lack clear interspecific morphological differences. This issue has led to considerable taxonomic confusion, since many of the intraspecific differences in the short-finned species seemingly exceed the interspecific differences (Brett and

Turner, 1983; Poeser, 2003). Already, in the original description of the complex, Hubbs (1926) described a number of races inseparable from *P. sphenops*, and was unable to delimit its distribution or characters. For some authors, all short-fins represent local variants from a highly polytypic and widespread species, *P. sphenops* (Valenciennes, 1846), ranging from the Río Grande drainage in north-eastern Mexico to coastal Venezuela (Rosen and Bailey, 1963). Another view, so far the most widely accepted, is that this short-fin group is formed by around 13 biological species with partially overlapping ranges and thus constitutes the *P. sphenops* species complex (Schultz and Miller, 1971; Miller, 2005) (Appendix A). Furthermore, some authors propose dividing this complex into two more complexes (Alpírez Quesada, 1971): a *P. sphenops* complex with species from the Pacific slope of Mexico through Central America and a *P. mexicana* (Steindachner, 1863) complex containing species from the Atlantic coast of Mexico to Nicaragua. Over their ranges, these confusing and morphologically variable species groups can be differentiated into two major dentitional types of tricuspid and unicuspid inner jaw teeth (Schultz and Miller, 1971) (Appendix A).

Furthermore, and again in contrast to cichlids, most species within sail and short-fin CA *Poecilia* do not occur sympatrically (Miller, 1966; Bussing, 1976). This pattern suggests that non-adaptive (i.e., little or no ecological or niche diversification, *sensu* Gittenberger, 1991) and morphostatic radiations (i.e., low morphological diversification, *sensu* Davis, 1993) might be occurring within this group. Thus, according to the allopatric distribution of these species we would expect predominantly vicariant-based cladogenetic patterns, coupled with post-speciation range expansions, rather than sympatric ecological adaptations (Rosen, 1975). This process would eventually give rise to geographically congruent evolutionary lineages and divergence times that agree with major geological events, as well as with patterns from other co-distributed organisms.

The genus *Poecilia* in general, and the *P. sphenops* complex in particular, have been subject of extensive studies, mostly related to the reproductive characteristics of the family (i.e., internal fertilization and live-bearing) (Pires et al., 2010; Meredith et al., 2011), and sexual selection and hybrid speciation (i.e., the case of *P. formosa*) (Hubbs and Hubbs, 1932; Avise et al., 1991; Stöck et al., 2010). Yet, its taxonomy and phylogeny are poorly understood (Rosen and Bailey, 1963; Miller, 1983; Rauchenberger, 1989; Lucinda, 2003; Lucinda and Reis, 2005). Thus far, morphological studies either conflict or have been unable to resolve the phylogenetic relationships of *Poecilia* and related genera or subgenera (Hubbs, 1933; Rosen and Bailey, 1963; Schultz and Miller, 1971; Miller, 1975, 1983; Rodríguez, 1997; Poeser, 2003). On the other hand, molecules have rendered better resolution (Brett and Turner, 1983; Ptacek and Breden, 1998; Breden et al., 1999), suggesting that the morphological characters used for phylogenetic inference might be homoplastic (Langerhans et al., 2005; Ornelas-García et al., 2008; Doadrio et al., 2009). Still, these studies either consider the *P. sphenops* complex within a wider taxonomic context (Hrbek et al., 2007; Meredith et al., 2011) or are based on just a few representatives from each taxonomic unit of the complex (Brett and Turner, 1983; Ptacek and Breden, 1998; Breden et al., 1999), therefore not considering any of the large geographical variation or hidden diversity of the group.

In this study we analyzed the patterns of genetic structure of the *P. sphenops* species complex, basing our hypotheses on a robust phylogenetic and geographic framework. For that purpose we sampled individuals across most of the distribution range of the *P. sphenops* species complex and obtained molecular information from mitochondrial and nuclear markers to answer the following questions: What is the pattern of species or lineages distribution? What are the main processes and events that shaped this

distribution and diversification? Is the current taxonomy of the group congruent with phylogenetic relationships?

2. Materials and methods

2.1. Sampling and DNA extraction, amplification and sequencing

Fish potentially belonging to the *Poecilia sphenops* species complex in CA (i.e., CA short-fin *Poecilia*) were collected by electrofishing or seining. Taxon sampling also included representatives of non-CA *Poecilia*: *P. caucana*, *P. hispaniolana*, which are distributed in northern SA and in the Island of Hispaniola, respectively; and from a closely related genus: *Limia vittata* and *L. melanonotata*, occurring in the Islands of Cuba and Hispaniola, respectively. We used the SA-distributed *P. reticulata* as outgroup, since morphological and molecular characters have proven it is closely related to *Micropoecilia* and sister to the *Poecilia*, *Limia* and *Pamphorychthis* group (Lucinda and Reis, 2005; Hrbek et al., 2007). (Fig. 1 and Table 1 summarize sample locations and specimens). Specimens were individually tagged with unique identification labels and small pieces of gill or muscle tissue were preserved in DMSO/EDTA buffer or 95% ethanol. DNA voucher specimens and their associated lots were subsequently preserved in buffered formalin and have been deposited in the Museo Nacional de Ciencias Naturales (CSIC), Spain and in the Neotropical Fish Collection at the Smithsonian Tropical Research Institute, Panama (Bermingham et al., 1997a) (Table 1). DNA was extracted from small pieces of tissue using QIA-GEN DNeasy Tissue kits.

The entire ATP synthase 6 and 8 (*ATPase 8/6*) genes were PCR amplified for all individuals using primers L8331 and H9236 (Meyer, 1993). In addition, a subset of the samples were PCR amplified for the NADH dehydrogenase subunit 2 (*ND2*) gene using primers: GLN and ASN (Kocher et al., 1995), the 5' region of the Cytochrome oxidase subunit I (*COI*) gene using primers: Fish-BCL and Fish-BCH (Baldwin et al., 2009), and the nuclear *S7* ribosomal protein (*S7*) with primers: *S7RPEX1F* and *S7RPEX3R* (Chow and Hazama, 1998).

All PCR reactions were performed in a final volume of 25 µl containing: 2.5 µl of 10X reaction buffer (50 mM KCl, 10 mM Tris-HCl), 2 mM MgCl₂, 0.4 µM of each primer, 0.2 mM of each dNTP, 20–50 ng of template DNA and 0.25 U of Amplitaq polymerase

(Applied Biosystems). The following thermocycler conditions were used for *ATPase 8/6* amplification: initial denaturation at 94 °C for 2 min, denaturation at 94 °C for 45 s, annealing at 53 °C for 45 s, extension at 72 °C for 90 s, repeated for 5 cycles, followed by 29 cycles at 94 °C for 45 s, 58 °C for 45 s and 72 °C for 90 s. Conditions for the remaining genes were: initial denaturation at 94 °C for 2 min followed by 30 cycles of 94 °C for 30 s, annealing at 50 °C (for *COI*), 52 °C (for *ND2*) or 60 °C (for *S7*) for 30 s, and extension at 72 °C for 60 s, and finally an extension step of 72 °C for 7 min. PCR products were purified using Exonuclease I and Shrimp Alkaline Phosphatase enzymatic reactions (USB Scientific) and sequenced in an ABI3130xl automated sequencer (Applied Biosystems) using the BigDye terminator v.3.1 kit (Applied Biosystems) with the same primers used for PCR. Chromatograms were checked and assembled using Sequencher v.4.8 (Gene Codes). Consensus sequences of the coding genes (*ATPase 8/6*, *COI* and *ND2*) were translated into aminoacids to check for stop codons, obvious sequencing errors, and for visual alignment, and subsequently untranslated into nucleotide sequences using Mesquite v.2.7.5 (Maddison and Maddison, 2011). Nuclear sequences (*S7*) were aligned with MAFFT v.6.818 (Katoh et al., 2002; Katoh and Toh, 2008) using the automatic algorithm, which selects the appropriate alignment strategy according to data size, the scoring matrix for nucleotide sequences was set to 1PAM/K = 2, the gap opening penalty was 1.53 and the offset value was 0.123. Alignments were confirmed with the software MUSCLE v.3.8.31 (Edgar, 2004) using a maximum of ten iterations and default parameters. Both alignments programs were run within the platform Geneious v.5.6.3 (Drummond et al., 2011).

2.2. Data analysis

Four datasets were constructed for analysis: one dataset containing all individuals sampled and sequenced for *ATPase 8/6*, a second dataset contained the subset of individuals sequenced for all the mtDNA coding genes (*ATPase 8/6*, *COI*, *ND2*), a third dataset included specimens sequenced for nuclear *S7* gene, and a fourth dataset included those individuals that were sequenced for all the mtDNA and nDNA genes. These data files were deposited in Dryad (doi:10.5061/dryad.fb280).

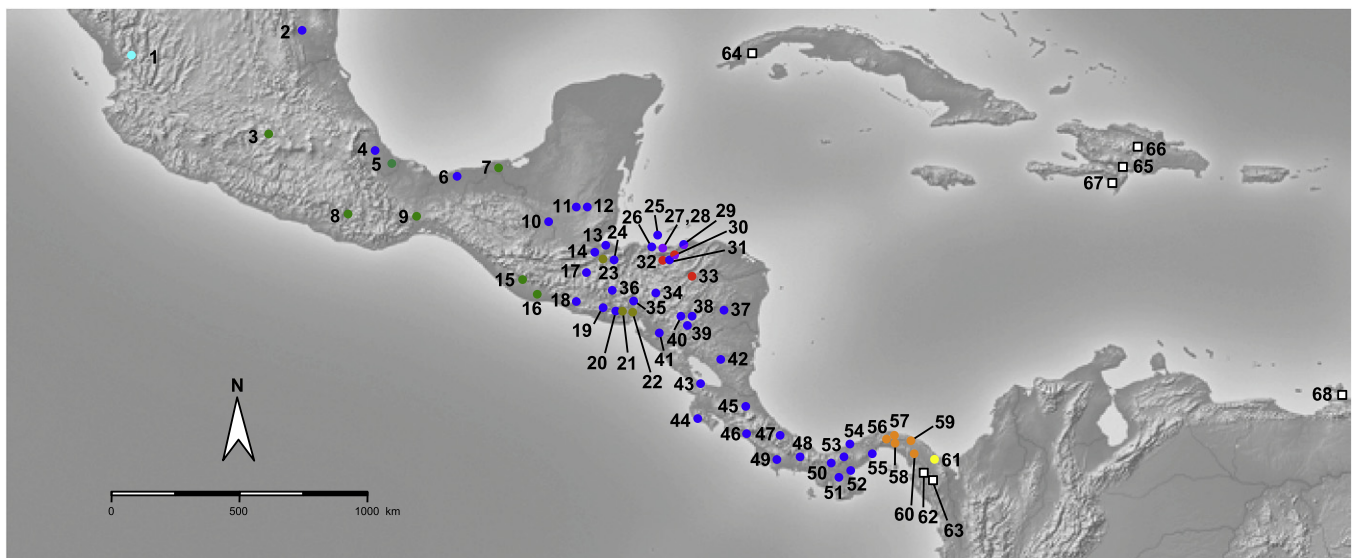


Fig. 1. Map of sampling localities in Central America. Filled circles represent CA short-fin *Poecilia* and empty squares are non-CA *Poecilia* and *Limia* species. Numbers of localities correspond to those indicated in Table 1 and colors correspond to the OTUs described based on the mtDNA analyses.

Table 1
Samples analyzed in this study, including information on the country, river drainage, slope and collection localities. Species name correspond to the morphological identification when possible, and mtDNA OTUs correspond to the assignment of each individual to the mitochondrial lineages. Genes amplified for each individual are indicated by their GenBank accession numbers. Locality numbers correspond to those indicated in Fig. 1.

Locality No.	Sample ID	Species	mtDNA OTU	River_Drainage	Locality	Country	Slope	ATPase 8/6	COI	ND2	S7
1	MEX-3800	<i>P. butleri</i>	<i>butleri</i>	Santiago	Tizatlán River	Mexico	Pacific	JX968561	JX968651	JX968697	JX968743
2	MEX-2880	<i>P. mexicana</i>	<i>mexicana</i>	Pánuco	Tamasopo River	Mexico	Atlantic	JX968562			
2	MEX-2880.2	<i>P. mexicana</i>	<i>mexicana</i>	Pánuco	Tamasopo River	Mexico	Atlantic	JX968563	JX968652	JX968698	JX968744
2	MEX-2881	<i>P. mexicana</i>	<i>mexicana</i>	Pánuco	Tamasopo River	Mexico	Atlantic	JX968564	JX968653	JX968699	JX968745
3	MEX-5011	<i>P. sphenops</i>	<i>sphenops</i>	Endorreic La Presa	Rodeo Pond	Mexico	Pacific	JX968565			
4	stri-x-3352	<i>P. mexicana</i>	<i>mexicana</i>	La Palma	La Palma River	Mexico	Atlantic	JX968566			
4	MEX-2349	<i>P. mexicana</i>	<i>mexicana</i>	La Palma	La Palma River	Mexico	Atlantic	JX968567			
5	MEX-2275	<i>P. catemacensis</i>	<i>catemacensis</i>	Lake Catemaco	Lake Catemaco	Mexico	Atlantic	JX968568	JX968654	JX968700	JX968746
5	MEX-2276	<i>P. catemacensis</i>	<i>catemacensis</i>	Lake Catemaco	Lake Catemaco	Mexico	Atlantic	JX968569	JX968655	JX968701	JX968747
6	MEX-2379	<i>P. sulphuraria</i>	<i>mexicana</i>	Grijalva-Usumacinta	Azufre River	Mexico	Atlantic	JX968570			JX968748
6	MEX-2380	<i>P. sulphuraria</i>	<i>mexicana</i>	Grijalva-Usumacinta	Azufre River	Mexico	Atlantic	JX968571	JX968656	JX968702	JX968749
7	MEX-2483	<i>P. mexicana</i>	<i>sphenops</i>	Grijalva-Usumacinta	Grande de Comitán River	Mexico	Atlantic	JX968572	JX968657	JX968703	
8	MEX-1107.1	<i>P. sphenops</i>	<i>sphenops</i>	Atoyac-Verde	Las Grutas River	Mexico	Pacific	JX968573			
8	MEX-1107.2	<i>P. sphenops</i>	<i>sphenops</i>	Atoyac-Verde	Las Grutas River	Mexico	Pacific	JX968574			
9	MEX-1633	<i>P. sphenops</i>	<i>sphenops</i>	Los Perros	Ojo de Agua	Mexico	Pacific	JX968575	JX968658	JX968704	
10	stri-7995	<i>P. sp.</i>	<i>mexicana</i>	Usumacinta	La Pasión River	Guatemala	Atlantic	JX968576			
11	stri-8033	<i>P. sp.</i>	<i>mexicana</i>	Lake Petén	Pond	Guatemala	Atlantic	JX968577			
12	stri-8084	<i>P. sp.</i>	<i>mexicana</i>	Belize	Sal Si Puedes Creek	Guatemala	Atlantic	JX968578	JX968659	JX968705	JX968750
13	GU-10231	<i>P. sphenops</i>	<i>mexicana</i>	Aguas Calientes	Aguas Calientes River	Guatemala	Atlantic	JX968579			
14	stri-8181	<i>P. sp.</i>	<i>mexicana</i>	Polochic	Amatillo River	Guatemala	Atlantic	JX968580			
14	stri-8185	<i>P. sp.</i>	<i>mexicana</i>	Polochic	Amatillo River	Guatemala	Atlantic	JX968581			
15	stri-7729	<i>P. sp.</i>	<i>sphenops</i>	Nahualate	Pachipa River	Guatemala	Pacific				JX968752
15	stri-7730	<i>P. sp.</i>	<i>sphenops</i>	Nahualate	Pachipa River	Guatemala	Pacific	JX968582	JX968660	JX968706	JX968753
15	stri-7731	<i>P. sp.</i>	<i>sphenops</i>	Nahualate	Pachipa River	Guatemala	Pacific	JX968583	JX968661	JX968707	JX968754
16	stri-7780	<i>P. sp.</i>	<i>sphenops</i>	María Linda	Sinacapa River	Guatemala	Pacific				JX968755
16	stri-7781	<i>P. sp.</i>	<i>sphenops</i>	María Linda	Sinacapa River	Guatemala	Pacific				JX968756
16	stri-7787	<i>P. sp.</i>	<i>sphenops</i>	María Linda	Sinacapa River	Guatemala	Pacific	JX968584	JX968662	JX968708	JX968757
17	SA-7	<i>P. sphenops</i>	<i>mexicana</i>	Lempa	Guajoyo River	El Salvador	Pacific	JX968585	JX968663	JX968709	JX968758
17	SA-9	<i>P. sphenops</i>	<i>mexicana</i>	Lempa	Guajoyo River	El Salvador	Pacific	JX968586			
18	SA-92	<i>P. sphenops</i>	<i>mexicana</i>	Olomega Pond	El Carmen	El Salvador	Pacific	JX968587			
18	SA-93	<i>P. sphenops</i>	<i>mexicana</i>	Olomega Pond	El Carmen	El Salvador	Pacific	JX968588			
19	SA-116	<i>P. sphenops</i>	<i>mexicana</i>	Olomega Pond	Coatepeque Lagoon	El Salvador	Pacific	JX968589			
20	SA-103	<i>P. sp.</i>	<i>mexicana</i>	Jocotal Pond	El Tránsito, San Miguel	El Salvador	Pacific	JX968590			
20	SA-104	<i>P. sphenops</i>	<i>mexicana</i>	Jocotal Pond	El Tránsito, San Miguel	El Salvador	Pacific	JX968591	JX968664	JX968710	JX968759
21	stri-8806	<i>P. sp.</i>	"sphenops" sp. 1	Goascorán	Goascorán River	Honduras	Pacific	JX968592	JX968665	JX968711	JX968760
22	stri-8859	<i>P. sp.</i>	"sphenops" sp. 1	Goascorán	Goascorán River	Honduras	Pacific				JX968761
22	stri-8823	<i>P. sp.</i>	"sphenops" sp. 1	Goascorán	Goascorán River	Honduras	Pacific	JX968593			
23	stri-8409	<i>P. hondurensis</i>	"sphenops" sp. 1	Ulúa	Naco River	Honduras	Atlantic	JX968594			
23	stri-8479	<i>P. hondurensis</i>	"sphenops" sp. 1	Ulúa	Camalote River	Honduras	Atlantic	JX968595			
24	stri-8411	<i>P. sp.</i>	<i>mexicana</i>	Ulúa	Naco River	Honduras	Atlantic	JX968596	JX968666	JX968712	JX968762
25	stri-4348	<i>P. orri</i>	<i>mexicana</i>	Jicotea Pond	Jicotea Pond	Honduras	Atlantic	JX968597			
26	stri-4308	<i>P. sphenops</i>	<i>mexicana</i>	Agua Buena	Agua Buena River	Honduras	Atlantic	JX968598	JX968667	JX968713	JX968763
27	stri-4414	<i>P. hondurensis</i>	<i>hondurensis</i>	Cangrejal	Cangrejal River	Honduras	Atlantic	JX968599			
28	stri-4323	<i>P. hondurensis</i>	<i>hondurensis</i>	Yaruca Blanco	Yaruca River	Honduras	Atlantic				JX968764
29	stri-8558	<i>P. sp.</i>	<i>mexicana</i>	Aguán	Taujica River	Honduras	Atlantic	JX968600			
29	stri-8565	<i>P. sp.</i>	<i>mexicana</i>	Aguán	Taujica River	Honduras	Atlantic	JX968601	JX968668	JX968714	JX968765
29	stri-8568	<i>P. hondurensis</i>	<i>hondurensis</i>	Aguán	Taujica River	Honduras	Atlantic				JX968766
29	stri-8534	<i>P. sp.</i>	<i>hondurensis</i>	Aguán	Chicho Creek	Honduras	Atlantic				JX968767
29	stri-8566	<i>P. hondurensis</i>	<i>hondurensis</i>	Aguán	Taujica River	Honduras	Atlantic	JX968602	JX968669	JX968715	JX968769
29	stri-8574	<i>P. sp.</i>	<i>hondurensis</i>	Aguán	Taujica River	Honduras	Atlantic	JX968603	JX968670	JX968716	JX968770
30	stri-8520	<i>P. hondurensis</i>	<i>hondurensis</i>	Aguán	Chicho Creek	Honduras	Atlantic	JX968604			
30	stri-8549	<i>P. sp.</i>	<i>orri</i>	Aguán	Chicho Creek	Honduras	Atlantic	JX968605			
31	stri-8607	<i>P. mexicana</i>	<i>mexicana</i>	Aguán	Medina River	Honduras	Atlantic	JX968606	JX968671	JX968717	JX968771

Table 1 (continued)

Locality No.	Sample ID	Species	mtDNA OTU	River_Drainage	Locality	Country	Slope	ATPase 8/6	COI	ND2	S7
32	stri-8747	<i>P. mexicana</i>	<i>orri</i>	Patuca	Tepemechin River	Honduras	Atlantic	JX968607	JX968672	JX968718	JX968772
33	stri-8706	<i>P. sp.</i>	<i>orri</i>	Patuca	Mercado Creek	Honduras	Atlantic	JX968608			
34	stri-8962	<i>P. mexicana</i>	<i>mexicana</i>	Choluteca	Agua Fria River	Honduras	Pacific	JX968609			
35	stri-8873	<i>P. sp.</i>	<i>mexicana</i>	Nacaome	Chiquito River	Honduras	Pacific	JX968610	JX968673	JX968719	JX968773
36	stri-8365	<i>P. gillii</i>	<i>mexicana</i>	Motagua	Higuito River	Honduras	Atlantic	JX968611			
37	stri-14256	<i>P. sp.</i>	<i>mexicana</i>	Prinzapolka	El Guineo River	Nicaragua	Atlantic	JX968612	JX968674	JX968720	JX968774
38	stri-13420	<i>P. gillii</i>	<i>mexicana</i>	San Juan	Viejo River	Nicaragua	Atlantic	JX968613			
39	stri-13327	<i>P. sphenops</i>	<i>mexicana</i>	Grande de Matagalpa	Grande de Matagalpa River	Nicaragua	Atlantic				JX968775
39	stri-13333	<i>P. hondurensis</i>	"sphenops" sp. 1	Grande de Matagalpa	Grande de Matagalpa River	Nicaragua	Atlantic				JX968776
39	stri-13328	<i>P. sphenops</i>	<i>mexicana</i>	Grande de Matagalpa	Grande de Matagalpa River	Nicaragua	Atlantic	JX968614	JX968675	JX968721	JX968777
39	stri-13330	<i>P. hondurensis</i>	"sphenops" sp. 1	Grande de Matagalpa	Grande de Matagalpa River	Nicaragua	Atlantic	JX968615	JX968676	JX968722	JX968778
40	stri-13876	<i>P. sp.</i>	<i>mexicana</i>	Coco	Venquilla Creek	Nicaragua	Atlantic				JX968779
40	stri-13887	<i>P. mexicana</i>	<i>mexicana</i>	Coco	Pantasma River	Nicaragua	Atlantic				JX968780
40	stri-13868	<i>P. sp.</i>	<i>mexicana</i>	Coco	Venquilla Creek	Nicaragua	Atlantic	JX968616			
40	stri-13869	<i>P. sp.</i>	<i>mexicana</i>	Coco	Venquilla Creek	Nicaragua	Atlantic	JX968617			
41	stri-13508	<i>P. sp.</i>	<i>mexicana</i>	Estero Real	Hato Grande River	Nicaragua	Pacific	JX968618			
42	stri-13666	<i>P. sp.</i>	<i>mexicana</i>	Escondido	Espavel	Nicaragua	Atlantic	JX968619			
43	stri-14722	<i>P. sp.</i>	<i>mexicana</i>	San Juan	Cárdenas River	Nicaragua	Atlantic	JX968620			
44	stri-1231	<i>P. gillii</i>	<i>mexicana</i>	Nosara	Nosara River	Costa Rica	Pacific	JX968621	JX968677	JX968723	JX968781
45	stri-1245	<i>P. gillii</i>	<i>mexicana</i>	San Juan	Sarapiquí River	Costa Rica	Atlantic	JX968622	JX968678	JX968724	JX968782
46	stri-2073	<i>P. gillii</i>	<i>mexicana</i>	Terraba	Peje River	Costa Rica	Pacific	JX968623			
46	stri-2074	<i>P. gillii</i>	<i>mexicana</i>	Terraba	Peje River	Costa Rica	Pacific	JX968624			
47	stri-4993	<i>P. gillii</i>	<i>mexicana</i>	Changuinola	Bongie River	Panama	Atlantic	JX968625			
48	stri-11626	<i>P. gillii</i>	<i>mexicana</i>	Guarumo	Canazas River	Panama	Atlantic	JX968626			
49	stri-112	<i>P. gillii</i>	<i>mexicana</i>	Chiriquí Viejo	Chiriquí Viejo River	Panama	Pacific	JX968627			
50	stri-9780	<i>P. gillii</i>	<i>mexicana</i>	Coclé del Norte	Moreno River	Panama	Atlantic	JX968628			
51	stri-3148	<i>P. gillii</i>	<i>mexicana</i>	Santa María	Lajas River	Panama	Pacific	JX968629			
52	stri-1118	<i>P. gillii</i>	<i>mexicana</i>	Antón	Antón River	Panama	Pacific	JX968630	JX968679	JX968725	JX968783
53	stri-15557	<i>P. gillii</i>	<i>mexicana</i>	Indio	Las Marías River	Panama	Pacific	JX968631			JX968784
54	stri-16781	<i>P. gillii</i>	<i>mexicana</i>	Lagarto	Mateo Creek	Panama	Atlantic	JX968632			
55	stri-15225	<i>P. gillii</i>	<i>mexicana</i>	Grande	Cárdenas River	Panama	Pacific	JX968633	JX968680	JX968726	JX968785
56	stri-16226	<i>P. gillii</i>	<i>gillii</i>	Chagres	Chagres River	Panama	Atlantic	JX968634	JX968681	JX968727	JX968786
57	stri-1320	<i>P. gillii</i>	<i>gillii</i>	Mandinga	Mandinga River	Panama	Atlantic	JX968635	JX968682	JX968728	
58	stri-11204	<i>P. gillii</i>	<i>gillii</i>	Bayano	Mamóní River	Panama	Pacific	JX968636	JX968683	JX968729	JX968787
59	stri-3706	<i>P. gillii</i>	<i>gillii</i>	Azúcar	Azúcar River	Panama	Atlantic	JX968637	JX968684	JX968730	JX968788
60	stri-3615	<i>P. gillii</i>	<i>gillii</i>	Bayano	Ipetí River	Panama	Pacific	JX968638	JX968685	JX968731	JX968789
61	stri-1736	<i>P. gillii</i>	"gillii" sp. 2	Acla	Acla River	Panama	Atlantic	JX968639	JX968686	JX968732	
61	stri-4162	<i>P. sp.</i>	"gillii" sp. 2	Acla	Creek at Acla River	Panama	Atlantic	JX968640	JX968687	JX968733	JX968790
62	stri-14905	<i>P. caucana</i>	<i>caucana</i>	Tuira	Santa Rosa River	Panama	Pacific	JX968641	JX968688	JX968734	JX968791
63	stri-6445	<i>P. caucana</i>	<i>caucana</i>	Tuira	Canglón River	Panama	Pacific	JX968642	JX968689	JX968735	JX968792
64	CU-371	<i>L. vittata</i>	<i>L. vittata</i>	La Fé	Pinar del Río	Cuba	Atlantic	JX968643	JX968690	JX968736	JX968793
64	CU-678	<i>L. vittata</i>	<i>L. vittata</i>	La Fé	Pinar del Río	Cuba	Atlantic	JX968644	JX968691	JX968737	JX968794
65	RD-243	<i>P. hispaniolae</i>	<i>P. hispaniolana</i>	Yaque del Sur	Mijo River	Dominican Republic	Atlantic	JX968645	JX968692	JX968738	JX968795
65	RD-244	<i>P. hispaniolae</i>	<i>P. hispaniolana</i>	Yaque del Sur	Mijo River	Dominican Republic	Atlantic	JX968646	JX968693	JX968739	JX968796
66	RD-36	<i>L. melanonotata</i>	<i>L. melanonotata</i>	Tabar	Tabar River	Dominican Republic	Atlantic	JX968647			
66	RD-76	<i>L. melanonotata</i>	<i>L. melanonotata</i>	Tabar	Tabar River	Dominican Republic	Atlantic	JX968648	JX968694	JX968740	JX968797
67	RD-121	<i>P. reticulata</i>	<i>P. reticulata</i>	Pedernales	Pedernales River	Dominican Republic	Atlantic	JX968649	JX968695	JX968741	JX968798
67	RD-122	<i>P. reticulata</i>	<i>P. reticulata</i>	Pedernales	Pedernales River	Dominican Republic	Atlantic	JX968650	JX968696	JX968742	JX968799

(continued on next page)

Table 1 (continued)

Locality No.	Sample ID	Species	mtDNA OTU	River_Drainage	Locality	Country	Slope	ATPase 8/6	COI	ND2	S7
68	stri-4289	<i>P. reticulata</i>	<i>P. reticulata</i>	Guanapo, Trinidad	Caroni River	Trinidad and Tobago	Atlantic	JX968561	JX968651	JX968697	JX968743
68	stri-4290	<i>P. reticulata</i>	<i>P. reticulata</i>	Guanapo, Trinidad	Caroni River	Trinidad and Tobago	Atlantic	JX968562			

The Akaike Information Criterion implemented in jModelTest v.0.1.1 (Posada, 2008) was used to determine the evolutionary model that best fits the data for each codon position when considering the ATPase 8/6 and mtDNA data sets, and for the S7 gene partition separately. The models selected were used for subsequent analyses.

2.2.1. Phylogenetic relationships

Phylogenetic relationships were inferred for each dataset separately, and in all cases each codon position was considered as a distinct data partition, as well as the S7 gene, when included. Bayesian inference (BI) was performed with MrBayes v.3.1.2 (Huelsenbeck and Ronquist, 2001). Two simultaneous Markov chain analyses (MCMC) were run for 5×10^6 generations each, to estimate the posterior probabilities distribution. Topologies were sampled every 500 generations and convergence between runs was assessed by monitoring the standard deviation of split frequencies in MrBayes and using the effective sampling size (ESS) criterion in Tracer v.1.5 (Rambaut and Drummond, 2007). A 50% majority-rule consensus tree was constructed after eliminating the first 5×10^5 generations (i.e. 1000 trees). Maximum Likelihood (ML) analyses were performed using Garli v.2.0 (Zwickl, 2006). To find the “best” ML tree, five replicates of each search were run from random starting trees generated by a stepwise addition algorithm and 50 attachments attempts per taxon. The maximum number of generations to be run to encounter a significantly better scoring topology (0.05) was set to 10,000. To estimate support, 1000 bootstrap replicates from one of the previous searches were run. In both phylogenetic analyses we considered nodes were highly supported when their posterior probabilities (Pp) were $\geq 99\%$ and their bootstrap values (Bv) $\geq 95\%$, and well supported when their Pp were $\geq 95\%$ and Bv $\geq 75\%$. The ATPase 8/6 BI tree including all the individuals analyzed in this study was deposited in TreeBASE (Accession number: 13188, <http://purl.org/phylo/treebase/phyloids/study/TB2:S13188>). Mean uncorrected *p*-distances were calculated within and among the main lineages recovered for the ATPase 8/6 genes using MEGA 5 (Tamura et al., 2011).

2.2.2. Gene tree discordances

To explore the alternative sources of discordance between mtDNA and nDNA topologies (retention of ancestral polymorphism vs. post-divergence gene flow) we used the posterior predictive checking approach of (Joly et al., 2009), implemented in JML v.1.0.1 (Joly, 2012). To do this, we first constructed a species tree using *BEAST (Drummond and Rambaut, 2007; Heled and Drummond, 2010). We defined 9 taxon sets, each corresponding to the main clades obtained from the analyses of the complete mtDNA dataset of the *P. sphenops* species complex (Supplementary Data 1), plus 5 taxon sets corresponding to the outgroup taxa. A strict clock model was assumed and the birth–death process was chosen as the coalescent prior for the species tree. Four different analyses were run for 4×10^7 generations each, sampling every 4000 generations. Then, we used the posterior distribution of species trees, population sizes and branch lengths resulting from *BEAST to simulate gene trees and sequence datasets under a coalescent model with no migration. The program JML calculates minimum pairwise

sequence distances between the simulated sequences of individuals including all possible comparisons, and tests for each dataset whether the observed minimum distance between sequences of two species is smaller than those expected under the simulated model that does not account for hybridization. If the observed distance is significantly smaller than the simulated distance, the null hypothesis that lineage sorting can explain the data is rejected and a hypothesis of hybridization might be accepted. Gene trees and sequences were simulated separately for the mtDNA and the nDNA dataset using the relative substitution rates across loci estimated by *BEAST. We discarded 25% of the 10,000 species trees recovered by *BEAST to simulate a total of 7500 gene trees for the mitochondrial and the nuclear genes partitions. Appropriate heredity scalars were selected for simulations of nuclear (2) and mitochondrial (0.5) markers, respectively. A significance level of 0.1 was specified in all runs.

2.2.3. Divergence time estimates

Finally, we estimated the time to the most recent common ancestor (TMRCA) and their credibility intervals (95% highest posterior density: HPD) for single representatives of the main lineages obtained from the analysis of the complete mtDNA and nDNA dataset. We used a relaxed molecular clock in BEAST v.1.6.2 (Drummond and Rambaut, 2007), with branch rates drawn from an uncorrelated lognormal distribution and a birth–death speciation model as a tree prior. Data was partitioned into four sets corresponding to the three codon positions of the mtDNA data set and the S7 nuclear gene. We placed a constraint on the age of specific nodes within the topology using time estimates for a known geologic event. We used the separation time between the islands of Hispaniola and Cuba (Wind Passage), here represented as the split between *L. melanonotata* and *L. vittata*. Different time estimates have been proposed for this event, and have been independently used in recent biogeographical studies of the Caribbean (Chakrabarty, 2006; Concheiro Pérez et al., 2007; Doadrio et al., 2009). For comparison, we performed the analysis using a recent estimate 17–14 Mya (Iturralde-Vinent, 2006) and a more ancient one 25–20 Mya (Pitman III et al., 1993). Four independent MCMC runs of 4×10^7 generations sampled every 4000 generations were combined using LogCombiner v.1.6.2. (<http://beast.bio.ed.ac.uk/LogCombiner>) and checked for convergence and adequate ESS (all posterior ESS ≥ 8000) using Tracer v.1.5. These analyses were repeated with and without outgroup (*P. reticulata*) to compare its effect on the estimated dates. Also, to be able to provide an estimate of the substitution rates for each of the genes analyzed depending the calibration points used, the molecular clock analyses were performed considering each gene as a separate partition and using the same models and priors described above.

3. Results

The complete ATPase 8/6 genes (168 and 684 bp, respectively) were sequenced for 86 ingroup and 4 outgroup individuals. A subset of 43 individuals representing the major lineages of the *P. sphenops* species complex in Central America and 3 outgroups were sequenced for the COI (651 bp) and ND2 (1047 bp) genes

Table 2

Dimensions, polymorphism and substitution models selected by jModeltest for each of the data sets analyzed.

	Sample size	Number of characters	Variable characters (%)	Parsimony informative characters (%)	Substitution model
ATPase 8/6 1st codon	90	284	77 (27.11%)	70 (24.65%)	TrN + G
ATPase 8/6 2nd codon	90	284	27 (9.51%)	21 (7.39%)	TrN + I
ATPase 8/6 3rd codon	90	284	202 (71.12%)	184 (64.79%)	TrN + I + G
mtDNA dataset 1st codon	46	850	188 (22.12%)	166 (19.53%)	GTR + G
mtDNA dataset 2nd codon	46	850	63 (7.41%)	49 (5.76%)	TrN + I + G
mtDNA dataset 3rd codon	46	850	619 (72.82%)	574 (67.53%)	TIM1 + I + G
S7	57	906	153 (16.89%)	115 (12.69%)	HKY + I

rendering a mtDNA data set of 2550 bp. Sequences of the nuclear S7 ribosomal protein gene (906 bp, including gaps) were obtained, without any ambiguous or heterozygous positions, from 57 specimens: 42 individuals already included in the mitochondrial data set and 15 additional individuals that were sequenced to increase the sample size at certain localities that could show evidence of hybridization or incomplete lineage sorting (Table 1). Therefore, the combined mitochondrial and nuclear DNA data set included 42 individuals and 3456 bp, of which 1012 were variable and 864 parsimony informative characters (Table 2 summarizes information for each of the codon and gene partitions and their substitution models). All sequences have been deposited in GenBank (accession numbers: JX968561–JX965799).

3.1. Phylogenetic relationships

Independent analysis of the codon-partitioned mitochondrial ATPase 8/6 genes and the mtDNA dataset (ATPase 8/6, ND2 and COI genes) yielded mostly congruent topologies of the phylogenies using ML and BI methods, with the exception of some unresolved relationships for the ATPase 8/6 gene dataset (Figs. 2 and 3). The mtDNA dataset rendered well-supported nodes for all the main clades in the phylogeny. Therefore, unless stated otherwise, support values in the text will be reported from the complete mtDNA data set.

Our results rejected the monophyly of *Poecilia*, since *P. hispaniolana* was the sister taxa of *Limia*. On the other hand, all CA *Poecilia* were monophyletic and indicate a sister relationship with SA *P. caucana* and all these with Antillean taxa (all Bv and Pp = 100%) (Figs. 2 and 3).

Within CA short-fin *Poecilia*, i.e. the *P. sphenops* species complex *sensu lato* (Schultz and Miller, 1971), two main clades were recovered with high bootstrap and posterior probability support (Bv = 100% and Pp = 100%) with a mean genetic distance of 7.8% (S.E. = 0.9% uncorrected *p*-distance for ATPase 8/6, Supplementary Data 1). Given this result, the two divergent groups of clades are treated hereafter as two distinct complexes following Alpírez Quesada (1971) (i.e., *P. sphenops* complex mtDNA clade and *P. mexicana* complex mtDNA clade). The former *P. sphenops* complex *sensu lato* (Schultz and Miller, 1971) will be called CA short-fin *Poecilia*. Operational Taxonomic Units (OTUs) are identified within each complex, and named according to the morphological field identification, geographic distribution and/or type locality. For clarity and brevity OTU names will be reported using their specific names, while the full species name will be used when referring to morphological species previously described.

The *P. sphenops* complex clade, recovered from the mtDNA data (Figs. 2 and 3), included individuals mainly distributed to the north of the Motagua–Polochic Fault, from Mexico to Honduras. Three OTUs were found: *catemacensis*, only occurring in Lake Catemaco in the Atlantic coast of Mexico, and two populations of *P. sphenops*, which were differentiated into *sphenops*, found in basins in the

Pacific versant of Mexico and Guatemala, and “*sphenops*” sp. 1, found in Atlantic and Pacific basins of Honduras. Phylogenetic relationships between these groups were not fully supported by any of the methods or datasets (Figs. 2 and 3).

The *P. mexicana* complex clade is distributed across all Central America in both slopes from Mexico to Panama. Five OTUs are found with significant phylogeographical structure (Bv > 75%, Pp > 90%): *butleri* in the Río Santiago basin in Mexico (only represented by one individual), *orri* in the Aguán and Patuca basins and *hondurensis* in the Aguán, Yaruca Blanco and Cangrejal basins in Honduras, *gillii* in the Río Chagres, Mandinga, Bayano and Azúcar, and “*gillii*” sp. 2 in the Río Acla in Panama (Figs. 2 and 3). In addition to these geographically well-delimited groups, a sixth large and widespread OTU was recovered (*n* = 49, Bv = 100%, Pp = 100%) including individuals from Mexico to Panama, most of which had been morphologically identified as *P. mexicana*, *P. sulphuraria*, *P. sphenops* and *P. gillii*, or were not assigned to any species due to ambiguous morphological characters (Table 1, Fig. 2). Because the type specimen of *P. mexicana* was collected in the Río Blanco (Veracruz, Mexico) close to our individuals from the Río Tamasopo we refer to this group as *mexicana* and call *gillii* to the OTU found in the Río Chagres in Panama, which is the type locality for this taxon. Groups within *mexicana* (mean *p*-distances = 1.2–2.3%, Supplementary Data 1) differed in their geographic distribution ranges, some of which were very restricted: for example groups I and II occurred only in Mexico, group III was found only in Costa Rica and group IV only in Nicaragua, while other groups spanned from Mexico to Panama (V) or from Nicaragua to Panama (VI) (Fig. 2).

Based on our complete mtDNA dataset, the widespread *mexicana* OTU showed a sister relationship with *butleri*, distributed in the Pacific basin of Mexico. Also, it is worth noting the highly supported relationships recovered for two OTU pairs in Honduras and Panama based on the topology for the mitochondrial data set: *orri* and *gillii*, and *hondurensis* and “*gillii*” sp. 2 (Table 3, Fig. 3).

The results obtained from the nuclear DNA data also supported the monophyly of CA short-fin *Poecilia* (Bv = 88%, Pp = 100%), which were differentiated into four clades (All Bv > 61%, Pp = 100%), although phylogenetic relationships among them were not resolved (Fig. 4). All mitochondrial OTUs were included in separate clades within the nDNA gene tree, except for *mexicana* which was not monophyletic, and was found in three out of the four nuclear clades. The first clade included all the individuals from the *P. sphenops* complex mtDNA clade: “*sphenops*” sp. 1 from Honduras constituted a separate sub-clade (Bv = 79%, Pp = 100%), whereas *sphenops* from Mexico and *catemacensis* grouped together, although with low support. Also, two individuals from Costa Rica that were grouped within *mexicana* according to their mtDNA, were found in this nuclear clade. The second clade was entirely formed by individuals from Honduras that formed the *hondurensis* OTU. The third clade included most of the geographically restricted OTUs found in the *P. mexicana* complex mtDNA clade: *butleri* from Mexico, *orri*

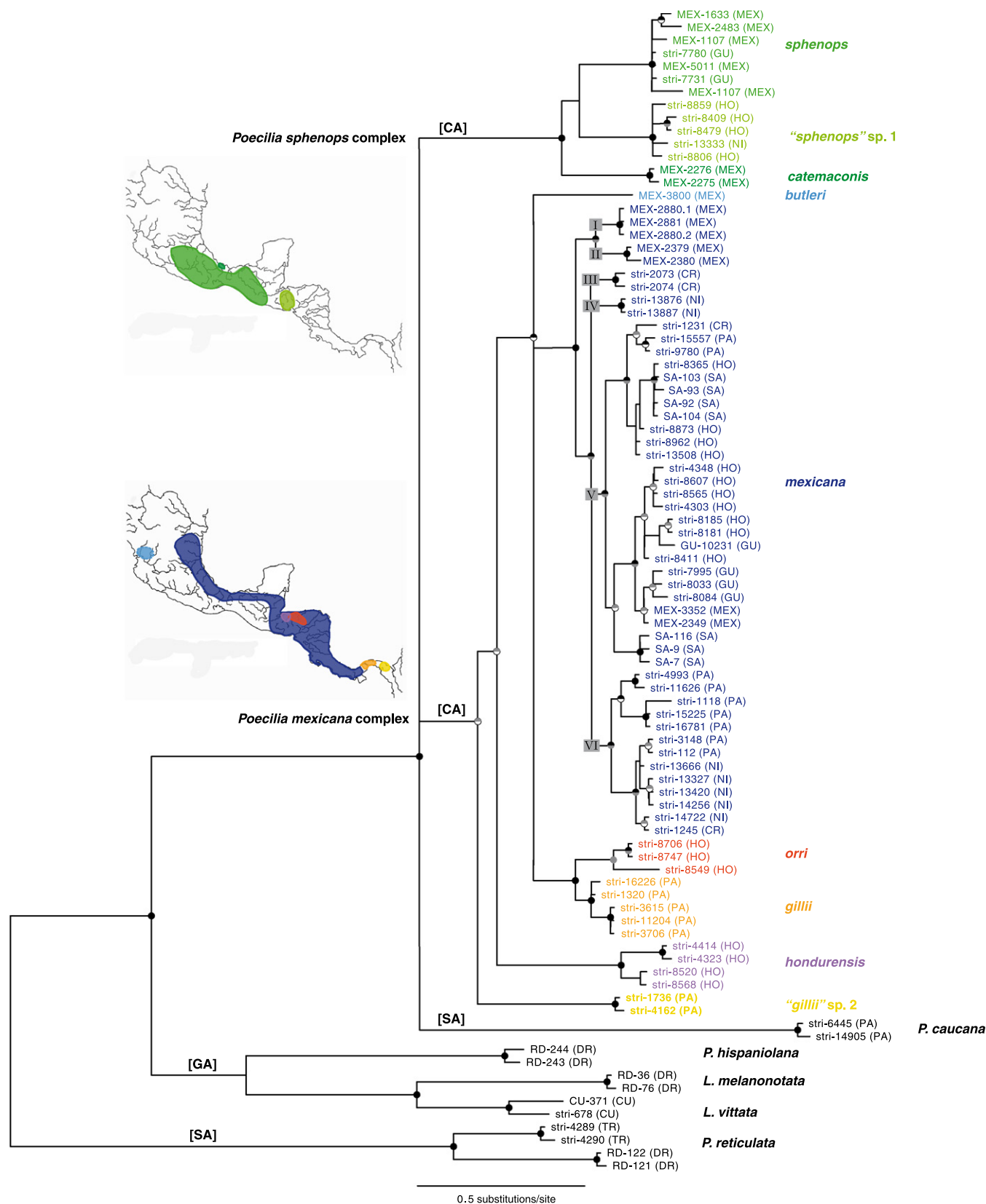


Fig. 2. Phylogenetic tree (50% majority rule consensus tree) rendered by Bayesian Inference analysis of the mitochondrial *ATPase 8/6* genes partitioned by codon position. Bullets on nodes indicate posterior probabilities (upper half) and bootstrap support (lower half) for the ML analysis. Black denotes highly supported nodes ($P_p \geq 99\%$, $B_v \geq 95\%$), gray denotes well supported nodes ($P_p \geq 95\%$, $B_v \geq 75\%$), and white or no bullet indicates no support. Sample codes as indicated in Table 1. Roman numerals indicate the six groups within the *mexicana* OTU (Supplementary Data 1). Letters between brackets above branches indicate the geographic distribution of each lineage: Central America (CA), South America (SA) and Greater Antilles (GA). Maps to the left schematically represent the distribution of the mtDNA OTUs.

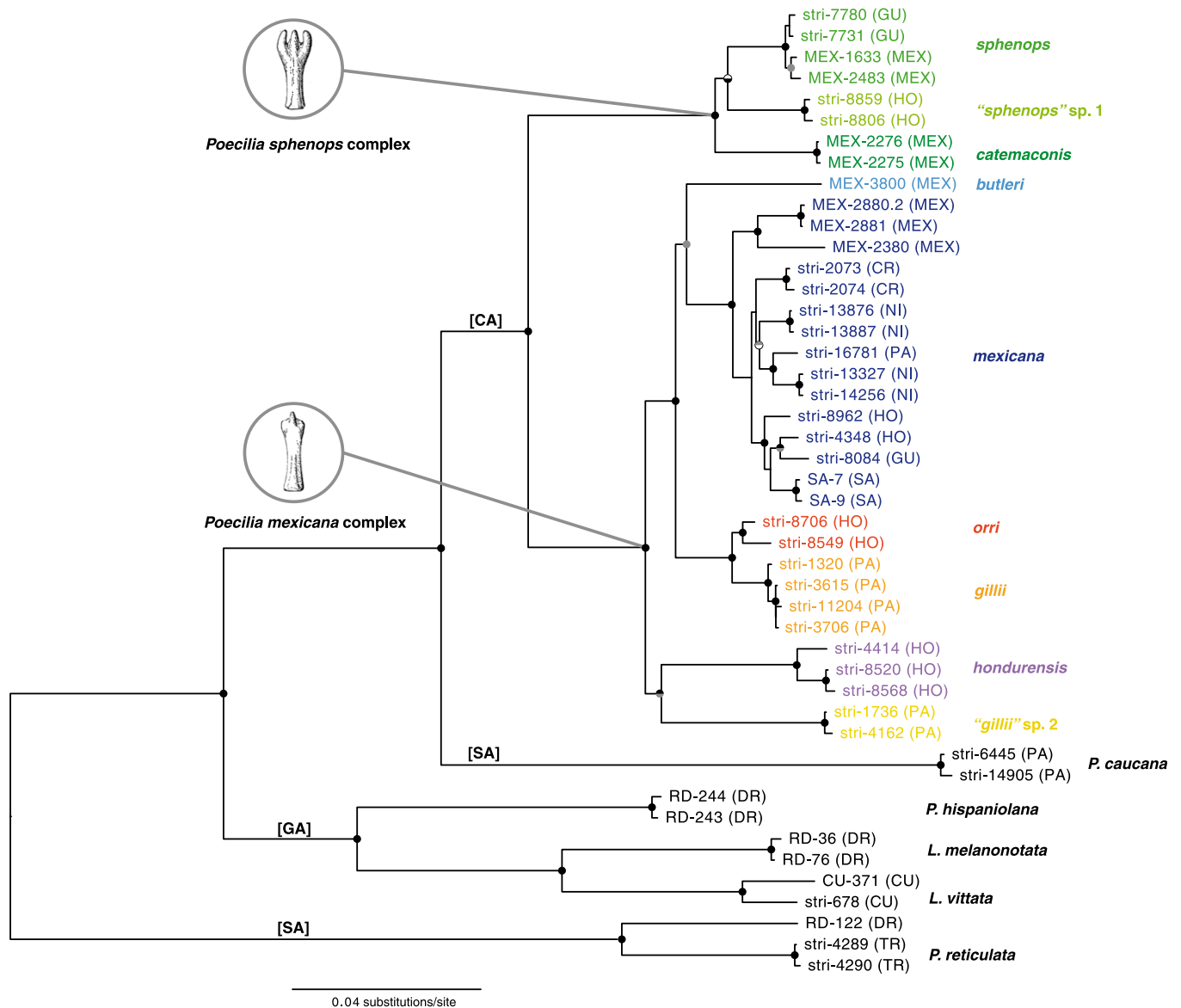


Fig. 3. Mitochondrial DNA tree. Phylogenetic tree (50% majority rule consensus tree) rendered by Bayesian analysis of the mitochondrial *ATPase 8/6*, *ND2* and *COI* genes partitioned by codon position. Bullets on nodes indicate posterior probabilities (upper half) and bootstrap support (lower half) for the ML analysis. Black denotes highly supported nodes ($P_p \geq 99\%$, $B_v \geq 95\%$), gray denotes well supported nodes ($P_p \geq 95\%$, $B_v \geq 75\%$), and white or no bullet indicates no support. The morphology of the inner jaw teeth for each lineage is indicated (drawings modified from Schultz and Miller, 1971). Letters between brackets above branches indicate the geographic distribution of each lineage: Central America (CA), South America (SA) and Greater Antilles (GA). Sample codes as indicated in Table 1.

Table 3

Uncorrected *p*-distances (in percentages) for *ATPase 8/6* genes between all mtDNA OTUs identified in Central American short-fin *Poecilia*. Values in the diagonal correspond to distances between individuals within OTUs. Numbers between parentheses are standard errors. Names correspond to those indicated in Fig. 2.

	<i>sphenops</i>	" <i>sphenops</i> " sp. 1	<i>catemaconis</i>	<i>mexicana</i>	<i>butleri</i>	<i>gillii</i>	<i>orri</i>	<i>hondurensis</i>	" <i>gillii</i> " sp. 2
<i>sphenops</i>	3.7 (0.4)								
" <i>sphenops</i> " sp. 1	4.6 (0.6)	0.4 (0.1)							
<i>catemaconis</i>	4.8 (0.6)	3.7 (0.7)	0.1 (0.1)						
<i>mexicana</i>	8.0 (0.9)	8.0 (1.0)	7.5 (1.0)	1.7 (0.2)					
<i>butleri</i>	8.7 (1.0)	8.9 (1.1)	7.9 (1.1)	4.0 (0.6)	N/A				
<i>gillii</i>	8.3 (1.0)	8.5 (1.2)	7.8 (1.1)	3.8 (0.6)	3.5 (0.6)	0.3 (0.1)			
<i>orri</i>	8.2 (1.0)	7.9 (1.1)	7.4 (1.0)	3.4 (0.5)	3.2 (0.6)	1.6 (0.4)	0.8 (0.2)		
<i>hondurensis</i>	8.4 (1.0)	8.4 (1.0)	8.4 (1.0)	5.2 (0.7)	5.5 (0.8)	5.4 (0.8)	5.0 (0.8)	0.8 (0.2)	
" <i>gillii</i> " sp. 2	8.5 (1.1)	8.0 (1.1)	8.8 (1.2)	5.6 (0.8)	5.8 (0.9)	5.7 (0.9)	5.4 (0.9)	5.7 (0.9)	0.1 (0.1)

from Honduras, *gillii* from Panama and "*gillii*" sp. 2 from Panama, in addition to individuals included in the *mexicana* OTU from Mexico to Panama. Finally, the fourth clade was entirely formed by speci-

mens of the mitochondrial *mexicana* clade from Mexico, which had been morphologically identified as *P. mexicana* and *P. sulphuraria*, and one individual from Honduras (Fig. 4).

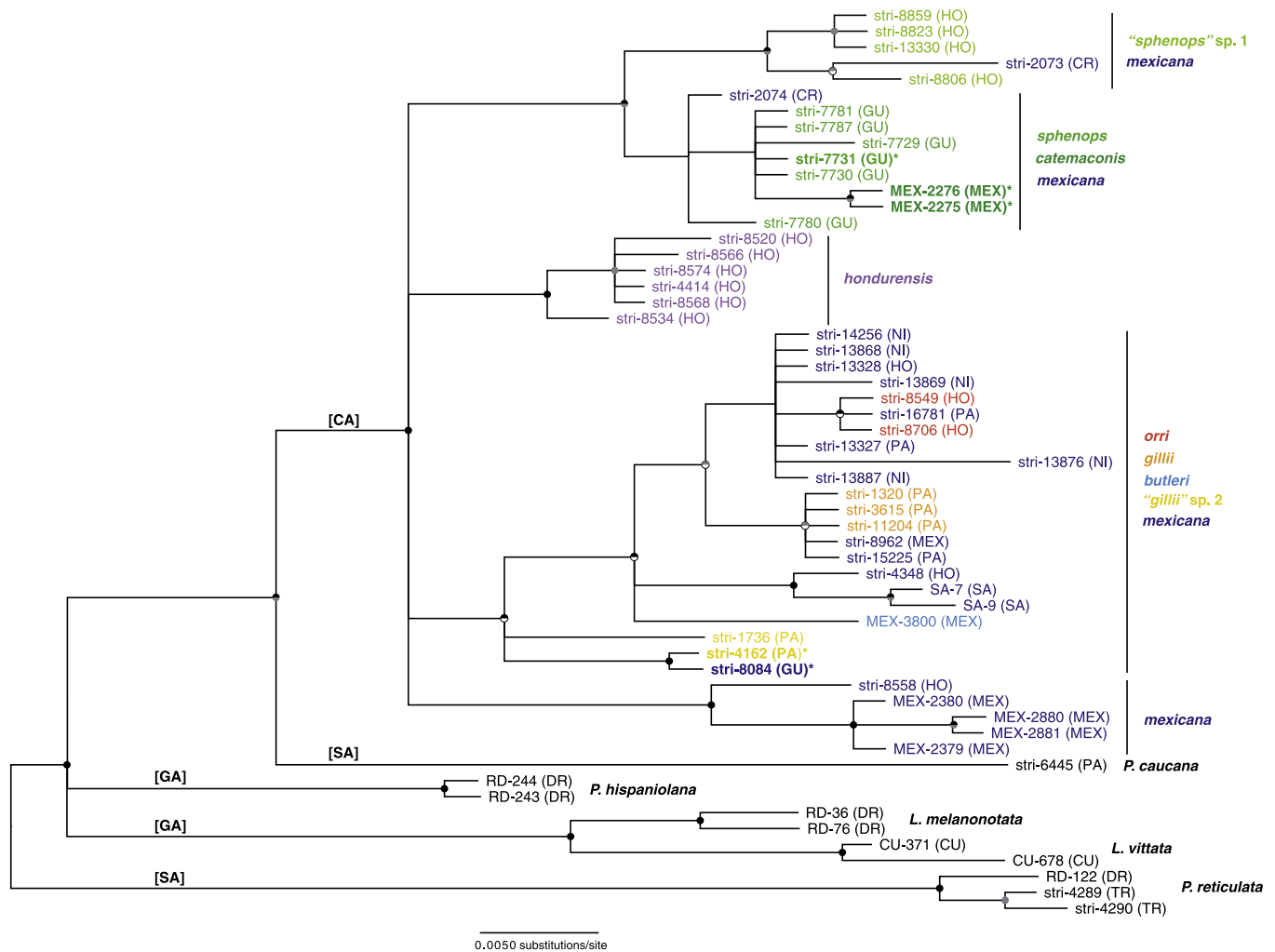


Fig. 4. Nuclear DNA tree. Phylogenetic tree (50% majority rule consensus tree) rendered by Bayesian analysis of the nuclear *S7* gene. Bullets on nodes indicate posterior probabilities (upper half) and bootstrap support (lower half) for the ML analysis. Black denotes highly supported nodes ($P_p \geq 99\%$, $B_v \geq 95\%$), gray denotes well supported nodes ($P_p \geq 95\%$, $B_v \geq 75\%$), and white or no bullet indicates no support. Letters between brackets above branches indicate the geographic distribution of each lineage: Central America (CA), South America (SA) and Greater Antilles (GA). Sample codes as indicated in Table 1. Taxa names with an asterisk are those with significantly small pairwise sequence distances, thus representing possible hybridization events, as obtained in JML.

The analysis of the combined mtDNA and nDNA datasets showed the same result as the combined mtDNA alone, and only some nodes within the *P. mexicana* complex clade showed lower support than the mtDNA tree. Thus, the phylogenetic position of *butleri* was not determined with confidence, and the sister relationship between *hondurensis* and “*gillii*” sp. 2 was only supported by the BI analysis. On the other hand, higher support values were obtained for the *P. sphenops* complex clade (All $B_v > 90\%$ and $P_p > 95\%$) (Supplementary Data 2).

3.2. Gene tree discordances

The analysis performed in JML showed contrasting results for the mtDNA and for the nDNA markers. In the first place, for the mtDNA markers, the test found no evidence of deviation from a coalescent model without hybridization in our data, indicating that incomplete lineage sorting alone can explain our data. On the other hand, when considering our nuclear marker, all pairwise distance comparisons were non-significant except for two species comparisons: *catemaconis* – *sphenops* and *mexicana* – “*gillii*” sp. 2 (Fig. 4), which observed pairwise distances had a significantly low probability of occurring under the simulated model ($P = 0.045$ and

$P = 0.047$, respectively). Therefore, a strictly bifurcating species tree model is not adequate to represent the relationships between these species, and consequently cannot reject the presence of hybridization. The species tree obtained with *BEAST is shown in Supplementary Data file 3.

3.3. Divergence time estimates

Our divergence time estimates indicated an old Eocene–Oligocene (between 54–26 Mya and 37–18 Mya depending on the calibration point considered) colonization of CA *Poecilia* (Fig. 5). Subsequently, the event that caused the divergence between the two main mitochondrial lineages of the *P. sphenops* and *P. mexicana* complexes was estimated to occur between 38–19 Mya and 26–13 Mya. Within each of these major lineages, the main radiation events leading to the appearance of the current OTUs occurred during the Miocene, between 23–6 Mya and 15–4 Mya. Finally, within OTU or population differentiation was estimated for both calibration points as younger than 5 Mya (Fig. 5). The analyses performed with and without outgroup sequences were highly concordant, and only slightly older estimates were obtained for the former (data not shown). The mean substitution rates estimated for each gene differed depending on the calibration point used, but showed

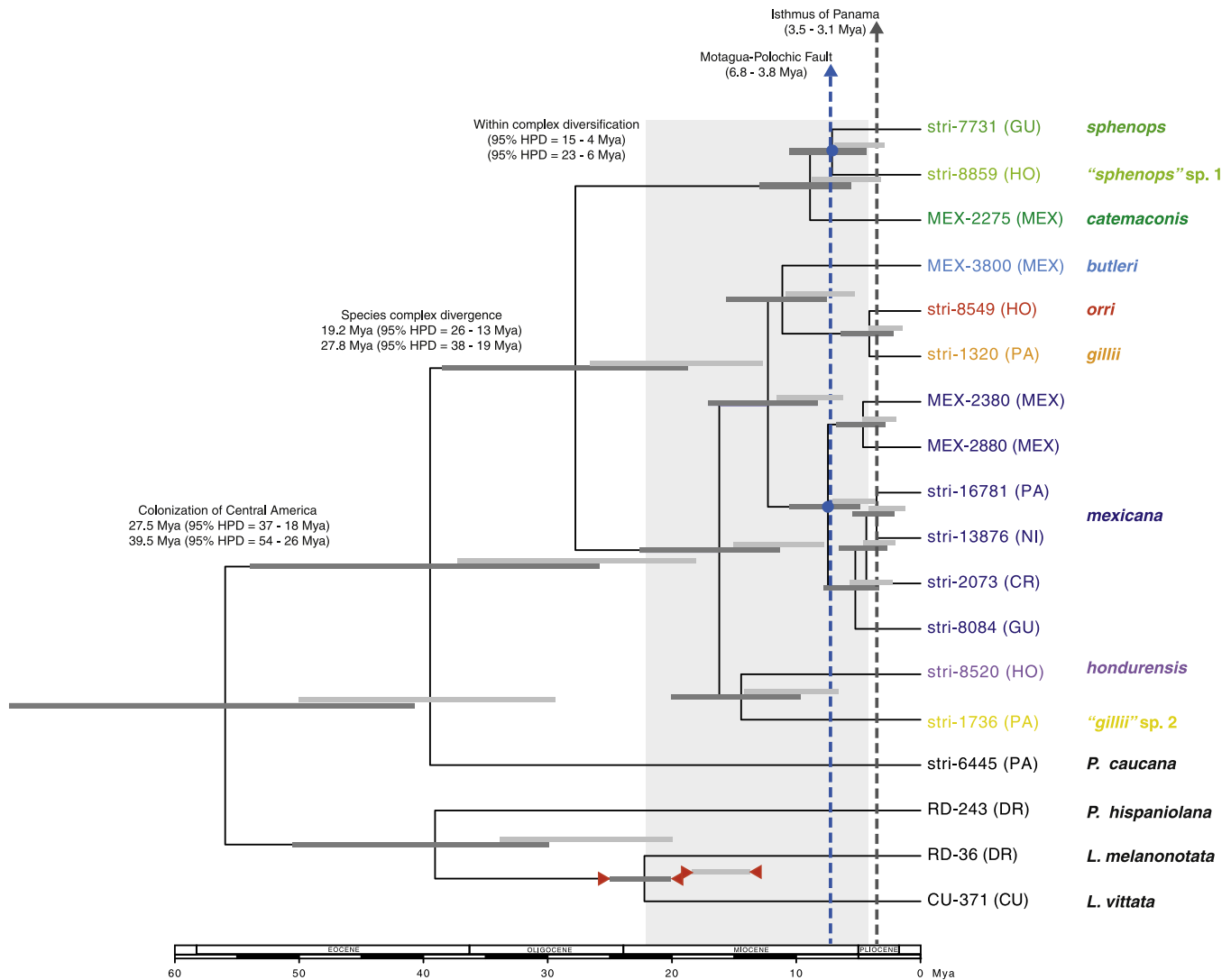


Fig. 5. Chronogram of CA short-fin *Poecilia* and related subgenera derived from a relaxed-clock Bayesian analysis based on the mtDNA data set and calibrated using the ancient estimate for the Cuba-Hispaniola split (25–20 Mya). Scale bar indicates time in My. Horizontal light gray bars are 95% HPD of the estimated TMRCA using a recent calibration for the Cuba-Hispaniola split (17–14 Mya) and dark gray bars using an ancient calibration (25–20 Mya) for the same event. Red triangles indicate upper and lower bounds of the employed calibration points. Mean estimates of TMRCA for the main divergence events and times for some biogeographic barriers discussed in the text are indicated. Also, the period encompassing the radiation events within each of the species complex is highlighted in gray. The dashed blue line indicates the Motagua-Polochic Fault and the dashed gray line indicates the closure of the Isthmus of Panama.

Table 4

Substitution rates estimated as substitution per site per million years for each of the gene sets analyzed according to two calibration points under a relaxed molecular clock using the software BEAST.

	Separation time between Hispaniola and Cuba			
	14–17 Mya		20–25 Mya	
	Mean	95% HPD	Mean	95% HPD
mtDNA	0.00294	(0.00216–0.00381)	0.00203	(0.00147–0.00262)
ATPase 8/6	0.00185	(0.00142–0.00232)	0.00268	(0.00297–0.00331)
COI	0.00146	(0.00108–0.00184)	0.00211	(0.00157–0.00265)
ND2	0.00221	(0.00172–0.00274)	0.00320	(0.00254–0.00394)
S7	0.00078	(0.00054–0.00105)	0.00054	(0.00037–0.00072)

overlapping HPD. The overall rates for the mtDNA data set were 0.00294 substitutions/site/My (s/s/My) (95% HPD: 0.00216–0.00381 s/s/My) and 0.00203 s/s/My (95% HPD: 0.00147–0.00262 s/s/My), for the “recent” and the “ancient” calibration points respectively. Considering each gene separately, the fastest

rate was in all cases estimated for the ND2 gene and the slowest for S7 (Table 4). All of our rate estimates were slower than those reported previously for other freshwater fishes which typically range from 0.0045 s/s/My for COI and 0.0054–0.0066 s/s/My for ATPase8/6, to 0.016–0.012 s/s/My for ND2 (Machordom and Doadrio, 2001; Sivasundar et al., 2001; Webb et al., 2004; reviewed in BurrIDGE et al., 2008).

4. Discussion

Morphological characters and taxonomy have been proven, in many cases, to be misleading and prevent inferences of fish evolutionary relationships (Martin and Bermingham, 1998; Perdices et al., 2002, 2005; Ornelas-García et al., 2008; Doadrio et al., 2009). In poeciliids, morphological characters have traditionally been used for systematic and biogeographical inference (Rosen and Bailey, 1963; Rosen, 1975; Bussing, 1976; Rauchenberger, 1988; Lucinda and Reis, 2005). However, these characters have failed to congruently resolve relationships below the genus level,

either because of a lack of reliable external diagnostic characters that do not offer much resolution at the species level (Hubbs, 1933; Alpírez Quesada, 1971; Schultz and Miller, 1971; Miller, 1983; Poeser, 2003), or as a consequence of the high homoplasy showed by some diagnostic characters, such as the gonopodium (Langerhans et al., 2005), all of which preclude the identification of species sinapomorphies. Therefore, in this study we used a robust approach based on molecular phylogenies to answer questions about the divergence patterns of CA short-fin *Poecilia*, i.e., the *P. sphenops* species complex *sensu lato*.

4.1. Phylogenetic relationships and lineage distribution

Our molecular phylogeny, based on both mtDNA and combined datasets, rejects the monophyly of *Poecilia*. *Poecilia hispaniolana* does not share its most recent common ancestor with CA *Poecilia*, but with the Antillean genus *Limia*, therefore further sampling on this group should be necessary to confirm its taxonomic status. On the other hand, all CA short-fin *Poecilia* were monophyletic, and within this group at least 9 mitochondrial OTUs can be described (Fig. 2). Two largely divergent mitochondrial lineages (mean *p*-distance = 7.8%, S.E. = 0.9%) were obtained that correspond well with the differentiation into two complexes (*P. sphenops* and *P. mexicana*) (Alpírez Quesada, 1971; Ptacek and Breden, 1998). Considering other poeciliids, the distance between species complexes is comparable to those observed, using other mitochondrial genes, between lineages of *Girardinus* (cytochrome *b* mean *p*-distance = 8.29%, S.E. = 0.9%), which are proposed to be different genera (Doadrio et al., 2009), or between species of *Poeciliopsis* (ND2 mean *p*-distance 9.34% – S.E. = 0.9%) (Mateos, 2005).

The two species complex showed also morphological differences based on two dentition types (Schultz and Miller, 1971). All members of the mitochondrial *P. sphenops* complex clade showed tricuspid inner teeth, whereas the mitochondrial *P. mexicana* complex clade showed unicuspid teeth. Most *Poecilia* species present unicuspid teeth, hence some authors consider this the ancestral state (Poeser, 1998). However, *Limia*, the sister genus of *Poecilia*, is characterized by tricuspid inner teeth (Rodríguez, 1997), so we could also hypothesize that this is the ancestral character from which unicuspid teeth evolved in the *P. mexicana* complex. In any case, and under both scenarios, our results support that teeth morphs evolved only once within each species complex. This contradicts a previous hypothesis proposing that teeth morphs evolved multiple times as a consequence of ecological character displacement following vicariance in at least three species pairs (Poeser, 1998).

Most of the drainages sampled contain only one mitochondrial lineage. This allopatric distribution of lineages suggests that divergence in allopatry following vicariance could be one of the main evolutionary forces promoting the diversification of CA short-fin *Poecilia*. Only three drainages: Aguán, Grande de Matagalpa and Ulúa, showed rivers with two sympatric OTUs (Table 1, Figs. 1 and 2), which, moreover, were not sister lineages, thus indicating a post-divergence river connection. All of these drainages are in the Atlantic slope of upper CA (Fig. 1), a pattern also observed in other neotropical freshwater fishes showing more sympatric mtDNA lineages in upper than in lower CA (Perdices et al., 2002; Ornelas-García et al., 2008). In the upper region, the continental shelf is very wide, probably promoting river anastomosis or drainage confluence during low sea level periods (Perdices et al., 2002; Smith and Bermingham, 2005), whereas in Panama, where the continental shelf is much narrower and steeper, a scenario of allopatrically distributed OTUs is observed.

4.2. Taxonomical implications

Although our results are highly consistent with previous molecular studies on the phylogenetic relationships among species within this group (Ptacek and Breden, 1998; Hrbek et al., 2007; Meredith et al., 2011), the increased sample size per taxon pinpoint other important aspects that differ with the current taxonomy of the group, particularly regarding the number of OTUs –or species– found and their distribution ranges. Therefore, based on our phylogenetic results, a systematic revision should be encouraged. For example, the distribution of the mitochondrial *mexicana* OTU, from the Pánuco Basin in Mexico down to Panama, was much larger than that described for the morphological species *P. mexicana* (Figs. 1 and 2, Appendix A) (Bussing, 2002; Miller, 2005). It is likely that the large morphological gradient and local differentiations have led to an over-classification of this complex, which is not supported by our molecular phylogeny.

On the other hand, *P. gillii* was previously described to occur roughly from Guatemala to Colombia (Bussing, 2002), largely overlapping with the distribution here observed for the *mexicana* OTU in upper and lower CA, but our data suggest that *gillii* is only restricted to rivers within the Chagres basin in Panama. Additionally, individuals from the Acla River within the proposed distribution of *P. gillii* in Panama constituted a largely differentiated lineage (>8.4% uncorrected *p*-distance compared to any other OTU, Table 3) and might constitute a different species. Indeed, specimens from the Gulf of Darien in the Panamanian Caribbean had already been described as a distinct taxon: *P. cuneata*; Garman, 1895 and *M. cuneata* (Meek and Hildebrand, 1916), and later synonymized with *P. gillii*, so a taxonomic revision is needed to determine if *P. cuneata* should be recovered. Interestingly, this OTU was recovered as a sister taxa of *hondurensis* in Honduras for the complete mtDNA data, as well as *gillii* was the sister taxon of *orri*, also in Honduras (Fig. 3). However, the large differences in branch lengths observed in the mtDNA tree, and consequently in their divergence time estimates (Fig. 5), as well as in ATPase 8/6 genetic distances between these pairs of OTUs (mean *p*-distance = 5.7%, S.E. = 0.9% and mean *p*-distance = 1.6%, S.E. = 0.4%, respectively) lead us to believe that this shared relationship is spurious rather than derived from a common event.

Finally, another OTU, “*sphenops*” sp. 1, possibly representing a new species, was described within the *P. sphenops* complex south to the Motagua–Polochic Fault in the Goascorán (Pacific) and Ulúa (Atlantic) basins in Honduras.

4.3. Hybridization or incomplete lineage sorting

Inconsistencies between gene trees are usually caused by introgressive hybridization (ancient or recent) or incomplete lineage sorting (Maddison, 1997). Incomplete lineage sorting arises when ancestral polymorphisms have not been completely sorted out by genetic drift in the descending lineages, resulting in non-monophyletic species (Joly, 2012); and introgressive hybridization is most commonly observed when two otherwise allopatric taxa contact and interbreed (Seehausen, 2004). Yet, teasing these processes apart is not easy, since both can generate very similar phylogenetic patterns (Joly et al., 2009; Martínez-Solano et al., 2012). The tests performed in JML indicated that for our nuclear marker, incomplete lineage sorting does not explain some of the phylogenetic relationships observed, thus hybridization cannot be ruled out (Fig. 4). However, significant comparisons never included sympatric individuals (Fig. 4), therefore we might deduce that the significantly small genetic distances between sequences, if due to hybridization, might be indicating an ancient introgression when taxa were distributed differently. Still, the rejection of the model of a bifurcating species tree without gene flow could also be due

to other processes such as gene duplication, substructure along the branches of the phylogeny, or parallel evolution (Joly, 2012).

Interestingly, only one of the significant comparisons included an individual of the mitochondrial *mexicana* OTU. Incomplete lineage sorting can thus be invoked as a major factor explaining the non-monophyly of this group in the nDNA gene tree (Fig. 4). Other evidence supporting incomplete lineage sorting is that the individuals of the mitochondrial *mexicana* that grouped within the nuclear *P. sphenops* complex clade presented normal unicuspid inner teeth, thus not showing any evidence of hybridization. Nevertheless, the question of why incomplete lineage sorting is only evident in the *mexicana* group remains unanswered. One possible explanation is that lineage sorting is a random process and consequently different markers would render different phylogenetic patterns (Joly, 2012). Another explanation could be deduced from the fact that *mexicana* showed the largest distribution range among the CA short-fin *Poecilia*. It is possible that both spatial and demographic expansions occurred in the past, resulting in vast distribution and increasing the effective population size, which would be inversely related to the time required to establish reciprocal monophyly (Hackett et al., 2008). In any case, further research regarding the demographic history of this group is being carried out to address this issue.

4.4. Historical biogeography

The monophyly recovered for CA *Poecilia* supports a single and ancient colonization event of CA. Additionally, the sister relationship of CA *Poecilia* with SA *P. caucana*, and then with Antillean *Limia* suggest that these genera experienced independent evolution and colonization events probably originating from SA (Hrbek et al., 2007), although further sampling is still needed to confirm this. The mean mtDNA sequence divergence between CA and SA *Poecilia* is 11–11.8%, representing a mean split time between 39.5 and 27.5 Mya (95% HPD ranging from 54 to 18 Mya). These dates are within the range of previous estimates for the timing of divergence between predominantly SA and non-SA poeciliids (43.91 Mya) and the colonization of *Poecilia* into CA (21.92 Mya) (Hrbek et al., 2007). If we assume that dispersal of *Poecilia* should be mostly over land, due to its mainly freshwater condition, this would imply that a land connection between South America and CA existed prior to these dates. However, our upper 95% HPD estimates predates by 10 million years the proposed Cretaceous connection between SA and NA between 144–65 Mya (Rosen, 1975, 1985) and Bussing's (1976) model of Late Cretaceous or Paleocene arrival of "old southern elements" via the hypothetical proto-Antillean land bridge (roughly ca. 75–65 Mya). On the other hand, poeciliids show a wide range of tolerance to saline environments, and therefore we cannot rule out that dispersal through marine environments could have occurred (Hrbek et al., 2007). In such case, the break-up between NA and SA would have not represented a barrier for these fishes as opposed to other primary freshwater fishes (Martin and Bermingham, 1998; Reeves and Bermingham, 2006). Some authors propose even older dates (Late Cretaceous – Early Paleogene) than our estimates for this continental break-up (Kerr et al., 1999; Iturralde-Vinent, 2006). Still, our estimates are much older than those proposed for *Rivulus* or heroine cichlids (15.9–18.4 Mya and 11.3–18 Mya, respectively) (Roe et al., 1997; Martin and Bermingham, 1998; Murphy et al., 1999; Concheiro Pérez et al., 2007). Therefore, this ancient colonization and long time of presence of *Poecilia* lineages in CA would have allowed for large range expansions and radiation before most primary freshwater fishes could be able to colonize from ca. 6 Mya to present (Bermingham and Martin, 1998).

The basal relationships of our phylogeny could also suggest an independent colonization of *Limia* from SA to the Antilles. Although

the data analyzed for this region is insufficient to draw well-supported conclusions, our results would be congruent in time with the GAARlandia hypothesis, which proposes a connection through a landbridge between the Greater Antilles and north western SA around 35–33 Mya (Iturralde-Vinent and MacPhee, 1999; Iturralde-Vinent, 2006; Hrbek et al., 2007). On the other hand, this hypothesis has been refuted for other freshwater fishes including cichlids and poeciliids, for which phylogenetic patterns are more congruent with vicariant scenarios proposing a common ancestry between Antillean and mainland Central America taxa (Chakrabarty, 2006; Doadrio et al., 2009; but see Murphy et al., 1999 and Concheiro Pérez et al., 2007).

After the invasion of CA and diversification of the short-fin and sail-fin *Poecilia* (not included in this study), the former group diverged into two lineages that might be recognized as different species complexes considering their large genetic variation and morphological synapomorphies. Following divergence, the main radiation events leading to current OTUs took place ca. 23–4 Mya (Miocene period). This dating is in agreement with the paleogeographic scenarios of lower CA that propose: (a) the existence of a peninsular extension of southern NA $\leq$ 16 Mya (Coates and Obando, 1996; Kirby and MacFadden, 2005), or (b) that between the middle and late Miocene (15–6 Mya) a shallow water connection between SA and CA existed (Coates and Obando, 1996), which together with the emerging Isthmus of Panama formed an island chain that might have been suitable for salinity-tolerant secondary freshwater fishes, such as poeciliids (Chervinski, 1994; Schlupp et al., 2002).

The *P. mexicana* complex showed an earlier and more geographically extended diversification than the *P. sphenops* complex. Also, regarding the short internodes in the terminal branches and long internal branches, it seems that the diversification process of *mexicana* has been fast but also very recent (ca. 5 Mya to present), as opposed to the early and rapid diversifications proposed for early colonizers of CA such as cichlids, *Rivulus* or *Rhamdia* (Martin and Bermingham, 1998; Murphy et al., 1999; Perdices et al., 2002). In our case the long internodes might indicate ancient long times of isolation due to vicariant events, or also large extinction of lineages. Furthermore, extinction of freshwater fishes due to marine incursions in the Nicaraguan depression could be an explanation for the patterns of sister-relationships between largely allopatric OTUs in Honduras and Panama. Multiple seaways have been proposed to exist along the Nicaraguan depression during the Pliocene, separating regions to the north and south (Coates and Obando, 1996). In other terrestrial taxa splitting times for this event have been estimated between 8.8 Mya and 4.1 Mya (Daza et al., 2010), which agree with the TMRCA between *orri* and *gillii* (95% HPD = 6.2–1.9 Mya) but not between *hondurensis* and "gillii" sp. 2 which TMRCA is much older (95% HPD = 20.4–6.4 Mya). After extinction, the *mexicana* lineage could have colonized the empty niche, leading to the observed phylogeographic pattern. On the other hand, the distribution of the *sphenops* complex is more restricted and basically delineates the limits of the Chortis block and Nuclear CA, which were the southernmost suitable habitats until the initial stages of the formation of lower CA ca. 15 Mya (Bussing, 1976; Coates and Obando, 1996; Iturralde-Vinent, 2006).

As it had been shown in previous studies in Costa Rica (Lee and Johnson, 2009), we did not find an overall correspondence of the phylogeographic breaks in CA short-fin *Poecilia* and the major ichthyological regions defined in CA (Bussing, 1976; Bermingham and Moritz, 1998; Smith and Bermingham, 2005). Nevertheless, we observed some shared phylogeographic patterns between the complexes here described and other freshwater fishes. In the first place, both complexes showed lineages separated by the Motagua–Polochic Fault (Fig. 5), which agree in time between them (11.4–3.5 Mya) and with other fish (7.8–6 Mya) (Ornelas-García

et al., 2008) and reptiles (8.8–4.1 Mya) (Daza et al., 2010). This fault represents the suture zone between the Maya and Chortis blocks and it has been largely recognized as an area of change and contact of many species lineages (Rosen, 1985; Perdices et al., 2002; Concheiro Pérez et al., 2007; Daza et al., 2010). In fact, in this region we found the only river (Naco River, Honduras) with both species complexes occurring in sympatry. Secondly, the break observed in western Panama separating *mexicana* and *gillii* OTUs has also been recovered for other species with low dispersal capabilities such as fishes and frogs (Perdices et al., 2002; Crawford

debted to O. Sanjur for her support during this project and to M. Iturralde-Vinent for discussion while preparing the manuscript. Finally, we wish to thank three anonymous reviewers that with their comments helped improve the manuscript. This study was partially financed by Project CGL2010-15231.BOS to I.D.

Appendix A

See Table A1.

Table A1

Systematic classifications proposed for the *P. sphenops* species complex, inner jaw tree morphology and proposed species distribution. Asterisks indicate species found in our study.

Species	Authors			Teeth morph.	Versant A: Atlantic, P: Pacific	Distribution ^{b,c,d,e}
	Rosen and Bailey (1963) ^f	Schultz and Miller (1971) ^g , Miller (2005) ^d	Alpírez Quesada (1971) ^a			
<i>P. sphenops</i> *	<i>P. sphenops</i> (one polytypic species)	<i>P. sphenops</i> species complex	<i>P. sphenops</i> complex	tricuspid	A, P	Atlantic slope of Mexico from the Palma Sola River to the Grijalva River basin, and in the Pacific slope from the Verde River basin into Guatemala
<i>P. chica</i>				tricuspid	P	Pacific slope of Mexico, in the basins of Cuetzamala River and Purificación in Jalisco
<i>P. marcellinoi</i>				tricuspid	P	El Salvador, Ilopango Lake basin
<i>P. maylandi</i>				tricuspid	P	Pacific slope of Mexico, Balsas River basin and Aguililla River
<i>P. butleri</i> *			<i>P. mexicana</i> complex	unicuspid	P	Pacific slope of Mexico to El Salvador
<i>P. catemacois</i> *				unicuspid	A	Mexico, Lake Catemaco
<i>P. gillii</i> *				unicuspid	A, P	Atlantic versant from Guatemala to Colombia, along the Pacific versant from Guatemala to the Terraba River in Costa Rica, and from the Grande to Bayano River in Panama
<i>P. hondurensis</i> *				unicuspid	A	Caribbean drainages of Honduras
<i>P. mexicana</i> *				unicuspid	A, P	Atlantic versant from northeastern Mexico to Costa Rica and in the Tamarindo River, in the Pacific slope of Nicaragua
<i>P. orri</i> *				unicuspid	A	Western coasts of Yucatán to northern Honduras
<i>P. salvatoris</i>				unicuspid	P	El Salvador
<i>P. sulphuraria</i> *				unicuspid	A	Baños del Azufre, Mexico
<i>P. teresae</i>				unicuspid	A	Belize

^a Alpírez Quesada (1971).

^b Bussing (2002).

^c Matamoros et al. (2009).

^d Miller (2005).

^e Poeser (2011).

^f Rosen and Bailey (1963).

^g Schultz and Miller (1971).

et al., 2007), and is hypothesized to be due to a narrow coastal corridor (Crawford et al., 2007).

In conclusion, our phylogenetic hypothesis suggests a single and very ancient colonization of CA *Poecilia*. The phylogeographic structure observed suggests long times of isolation due to vicariant events followed by recent expansions, probably facilitated by the physiological tolerance to salinity, during the initial stages of the formation of the isthmus of Panama.

Acknowledgments

We are grateful to all the people who assisted in the collection of specimens: P. Garzón-Heydt, F. Morcillo, A. Perdices, M. Pomes and A. de Sostoa. We thank L. Alcaraz for assisting in the laboratory and S. Perea for so many insightful comments while preparing this manuscript. We also thank W. Matamoros for his help identifying specimens and suggestions, I. Martínez-Solano for helpful discussion and the members of the Bermingham lab for critical reading and English revision of the text, especially to: I. Cerón-Souza, S. Lipshutz, S. Picq, O. Puebla, C. Salazar and K. Saltonstall. We are in-

Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.ympev.2012.12.012>.

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